

Causal Inference Methods for Infectious Disease Prevention Trials

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Introduction: Causal Inference to Study Infectious Disease

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Outline

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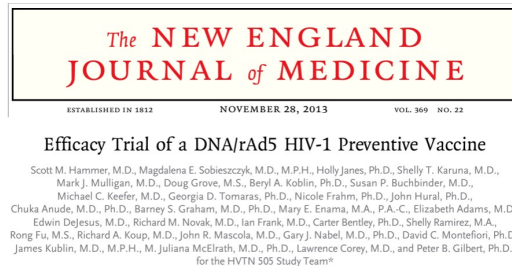
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- **Project 3:** Causal Inference from Cluster Randomized Trials with Differential Nonresponse

Project 1 : Addressing Confounding and Continuous Exposure Measurement Error Using Corrected Score Functions

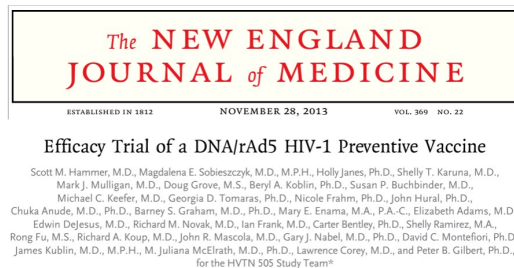
Brian Richardson, Brian Blette, Peter Gilbert, Michael Hudgens (2025)

Motivation: HVTN 505 trial



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- Stopped early after reaching predetermined cutoffs for efficacy futility ([Hammer et al., 2013](#))

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The Journal of Infectious Diseases

MAJOR ARTICLE



Higher T-Cell Responses Induced by DNA/rAd5 HIV-1 Preventive Vaccine Are Associated With Lower HIV-1 Infection Risk in an Efficacy Trial

Holly E. Janes,¹ Kristen W. Cohen,² Nicole Frahm,³ Stephen C. De Rosa,¹ Brittany Sanchez,¹ John Hural,¹ Craig A. Magaret,¹ Shelly Karuna,¹ Carter Bentley,¹ Raphael Gottardo,¹ Greg Finak,¹ Douglas Grove,² Mingchao Shen,¹ Barney S. Graham,³ Richard A. Koup,³ Mark J. Mulligan,⁴ Beryl Koblin,⁵ Susan P. Buchbinder,⁶ Michael C. Keefer,⁷ Elizabeth Adams,⁸ Chuka Anude,^{8a} Lawrence Corey,¹ Magdalena Sobieszczyk,^{1b} Scott M. Hammer,^{1b} Peter B. Gilbert,¹ and M. Juliana McElrath¹

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What Do We Mean by Causal Effect?

“What would be the risk of HIV if, possibly counter to fact, somebody were to have biomarker level **a**?”
([Neyman et al., 1935](#); [Rubin, 1974](#); [Holland, 1986](#); [Hernan and Robins, 2024](#))

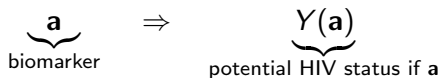
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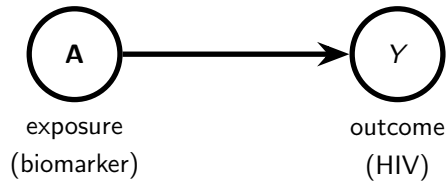
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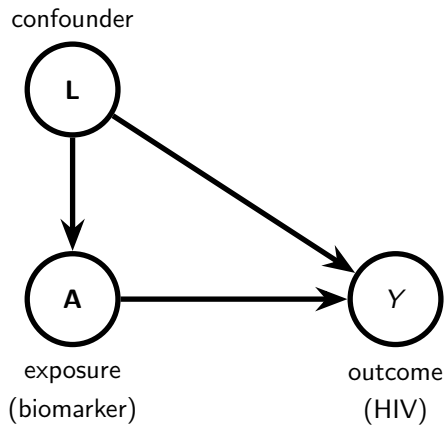
$$\underbrace{\mathbf{a}}_{\text{biomarker}} \Rightarrow \underbrace{Y(\mathbf{a})}_{\text{potential HIV status if } \mathbf{a}} \Rightarrow \underbrace{E\{Y(\mathbf{a})\}}_{\text{potential HIV risk if } \mathbf{a}}$$

Estimand (**dose-response surface**): $\eta(\mathbf{a}) \equiv E\{Y(\mathbf{a})\}$ for $\mathbf{a} \in \mathcal{A}$

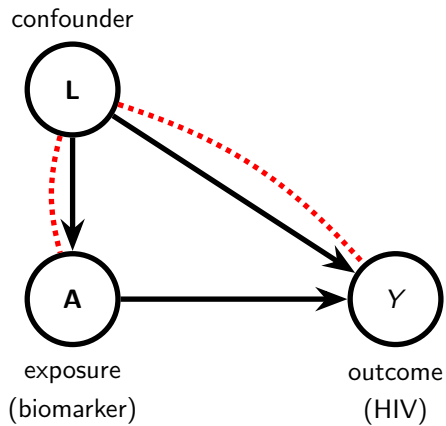
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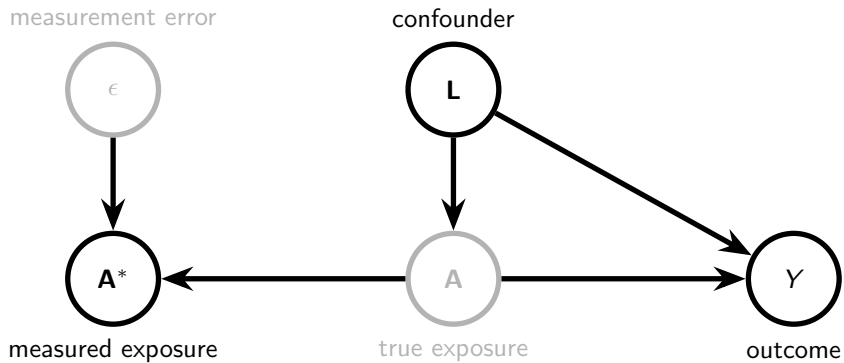
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- These can all be framed as **M-estimators** ([Stefanski and Boos, 2002](#))

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Score Function: a function of the observed data and the parameter of interest

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$\hat{\theta}$ is consistent and asymptotically normal and has a simple variance estimator

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This can be expressed as an M-estimator with score function

$$\Psi_{0-GF}(Y, \mathbf{L}, \mathbf{A}; \theta_{GF}) = \begin{bmatrix} \{Y - \mu(\mathbf{L}, \mathbf{A}; \beta)\} \partial_{\beta} \mu(\mathbf{L}, \mathbf{A}; \beta) \\ \eta(\mathbf{a}) - \mu(\mathbf{L}, \mathbf{a}; \beta) \end{bmatrix}$$

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Doubly robust* to models for $\mu(\mathbf{L}, \mathbf{A}; \beta)$ and $f_{\mathbf{A}|\mathbf{L}}(\mathbf{A}|\mathbf{L})$

Addressing Confounding and Measurement Error

Can we just substitute $\mathbf{A}^*$ for $\mathbf{A}$ and find the solution to

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No! This leads to bias in $\hat{\theta}$ because

$$E\{\psi_0(Y, \mathbf{L}, \mathbf{A}^*; \theta_0)\} \neq \mathbf{0}.$$

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We need a new score function ψ_{CS} such that

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Given the “oracle” score function Ψ_0 , the “corrected score” function Ψ_{CS} can be created following Novick and Stefanski (2002) ([Novick and Stefanski, 2002](#)):

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- 3 Take the expectation over the additional measurement error $\tilde{\epsilon}$.

Corrected Score Functions (contd)

Under certain conditions, the corrected score function Ψ_{CS} is then unbiased, meaning

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The G-Formula, IPW, and DR score functions all satisfy these conditions, and so can be “corrected:”

$$\Psi_{0-GF} \longrightarrow \Psi_{CS-GF}$$

$$\Psi_{0-IPW} \longrightarrow \Psi_{CS-IPW}$$

$$\Psi_{0-DR} \longrightarrow \Psi_{CS-DR}$$

Monte Carlo Corrected Score Functions

- Sometimes we can find a closed-form algebraic expression for

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$$\begin{aligned}\mathbb{E}\{f(\tilde{\epsilon})\} &\approx B^{-1} \sum_{b=1}^B f(\tilde{\epsilon}_b) \\ \implies \Psi_{MCCS}^B(Y, \mathbf{L}, \mathbf{A}^*; \theta) &= B^{-1} \sum_{b=1}^B \operatorname{Re}\{\Psi_0(Y, \mathbf{L}, \mathbf{A}^* + i\tilde{\epsilon}_b; \theta)\}\end{aligned}$$

Simulation Setting

$$L \sim \mathcal{N}(0, 0.36)$$



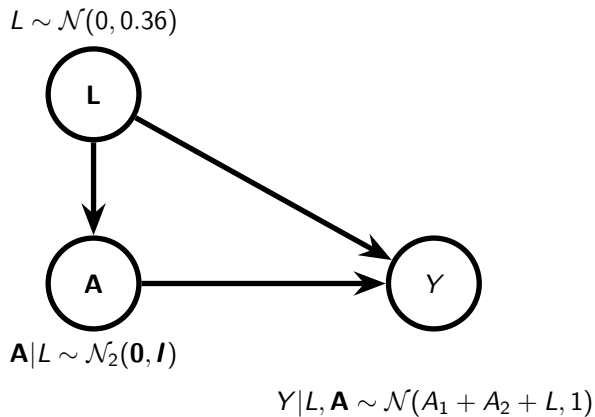
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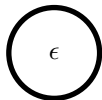
$$\mathbf{A}|L \sim \mathcal{N}_2(\mathbf{0}, I)$$

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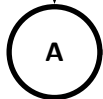


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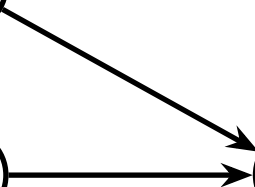
$$\epsilon \sim \mathcal{N}_2(\mathbf{0}, \sigma_{me}^2 \mathbf{I})$$



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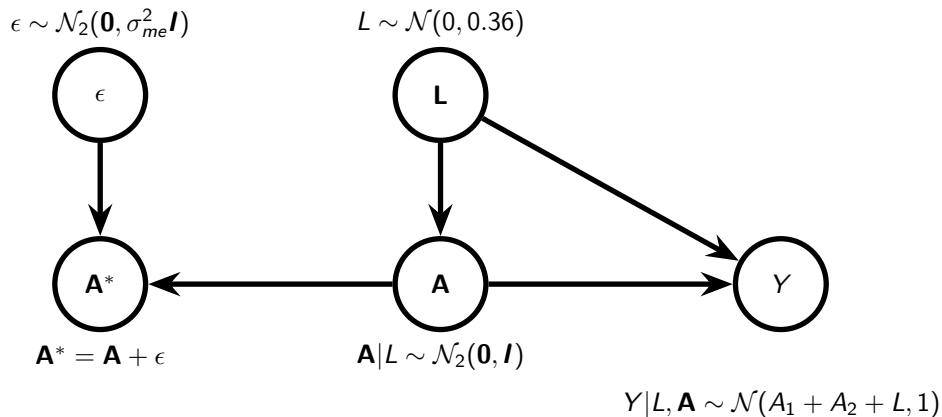


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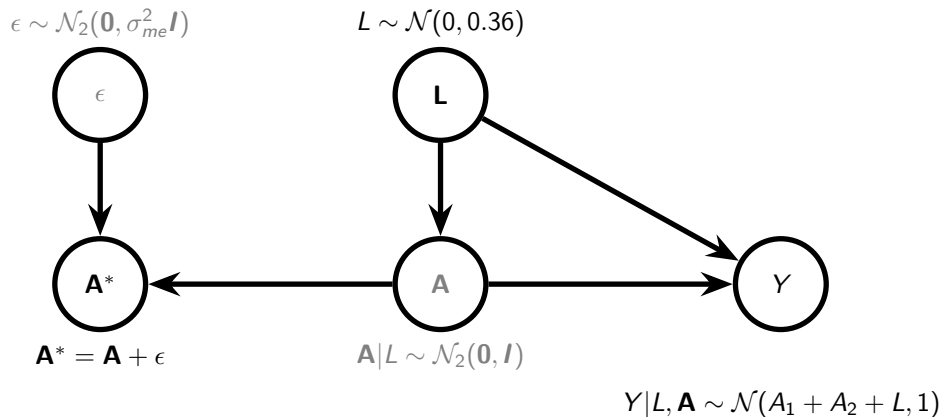


$$Y|L, \mathbf{A} \sim \mathcal{N}(A_1 + A_2 + L, 1)$$

Simulation Setting



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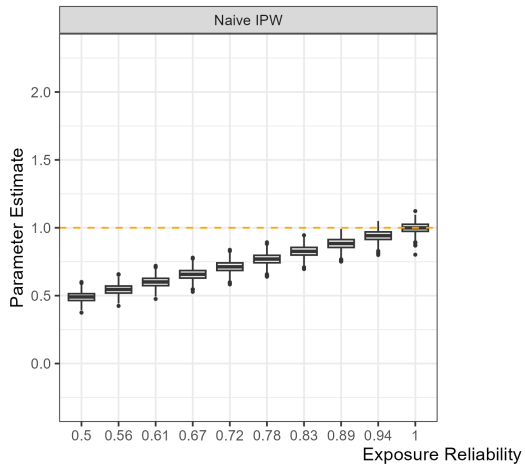
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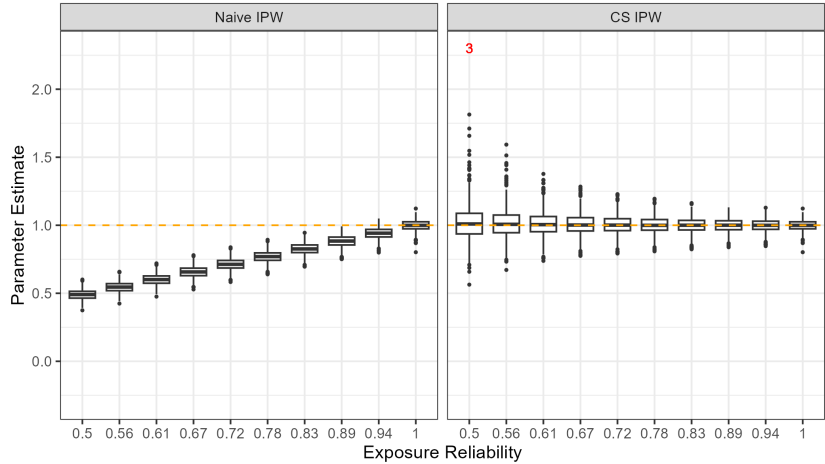
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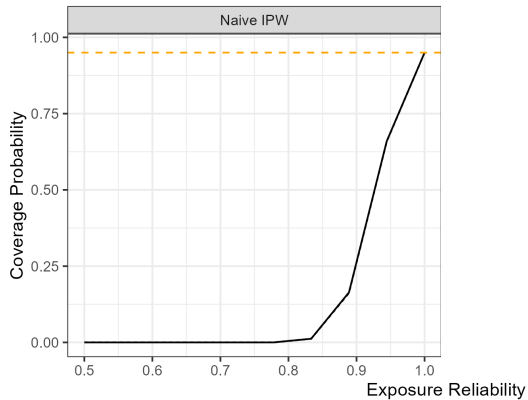
Simulation Results: Estimator



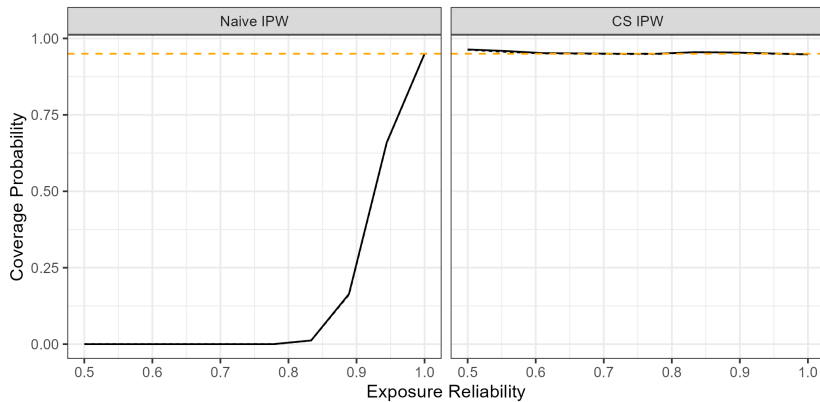
Simulation Results: Estimator



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Application: HVTN 505 Trial

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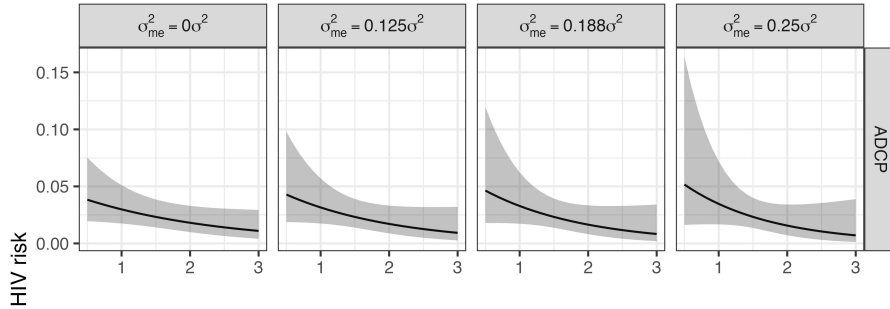
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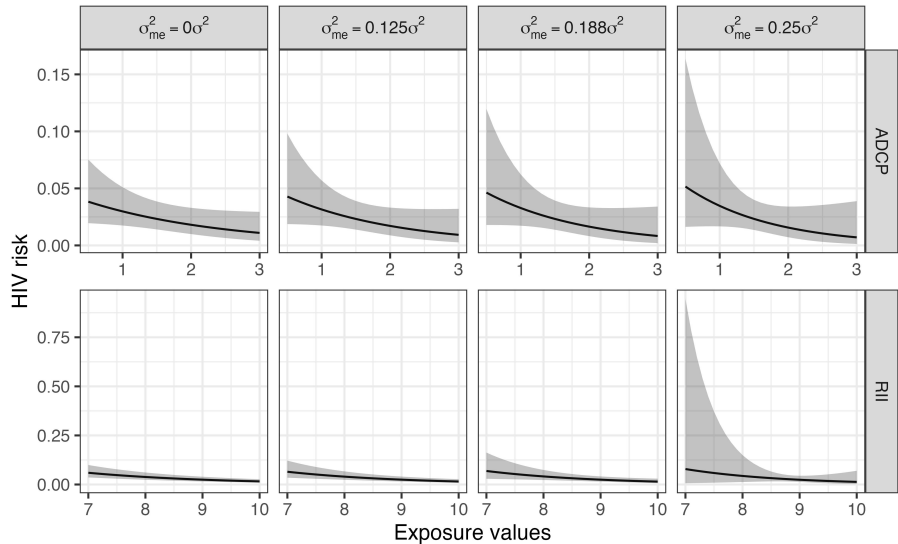
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Application: HVTN 505 Trial



Exposure values

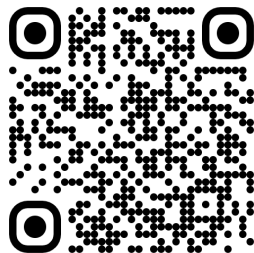
Application: HVTN 505 Trial



Summary of Project 1



Paper in *Biometrics* (2025)



GitHub R package

Mismex: Causal Inference for Mismeasured Exposures

Project 2: Social Distancing to Reduce Transmission of Influenza-Like-Illness on College Campuses: the eX-FLU Trial

Brian Richardson, Allison Aiello, Michael Hudgens

eX-FLU Trial

- **eX-FLU**: trial to evaluate a social distancing intervention on a college campus during flu season ([Aiello et al., 2016](#); [Zivich et al., 2020](#))

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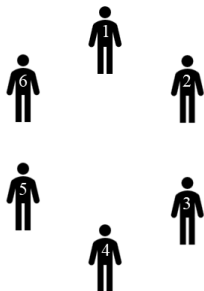
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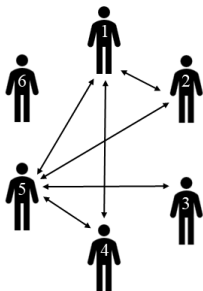
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- **Central question**: does the encouragement-to-isolate intervention reduce transmission of ILI?

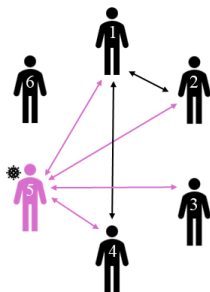
Transmission of Influenza-Like-Illness



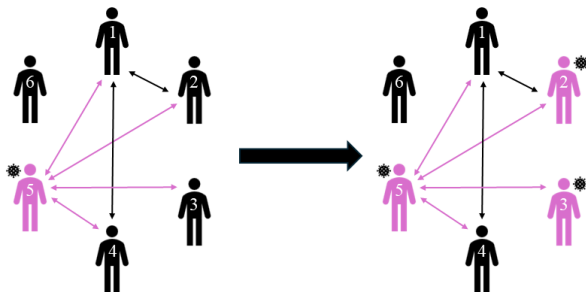
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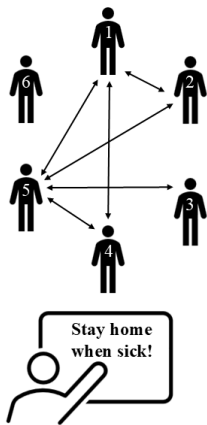
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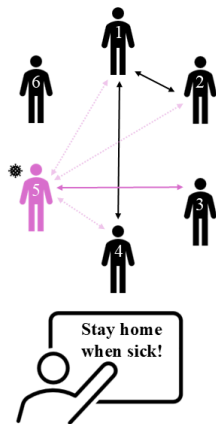
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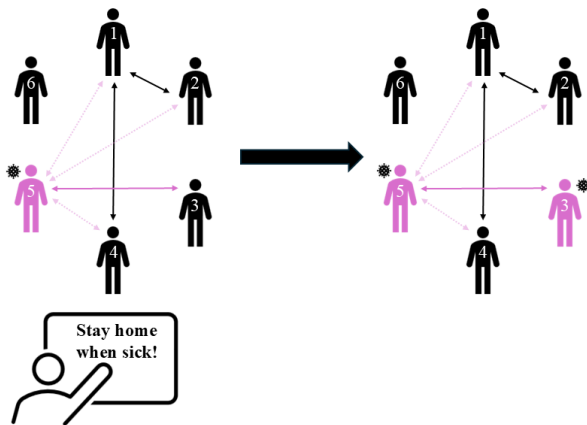
Example I: Intervention Affects Network and Transmission



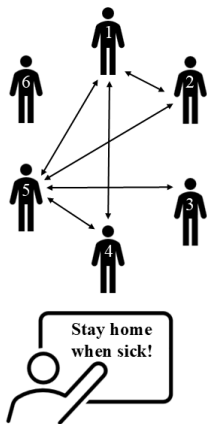
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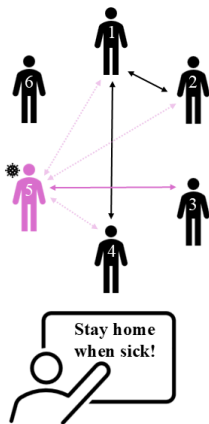
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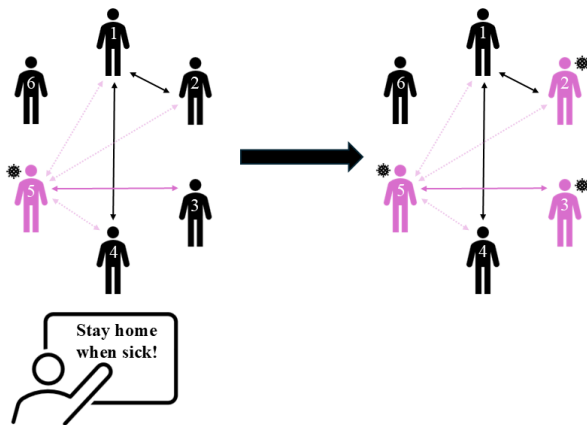
Example II: Intervention Affects Network, Not Transmission



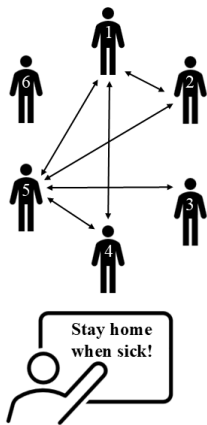
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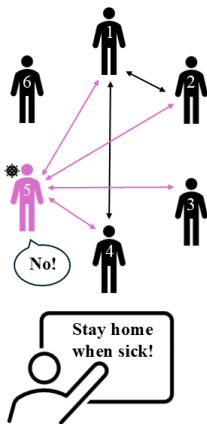
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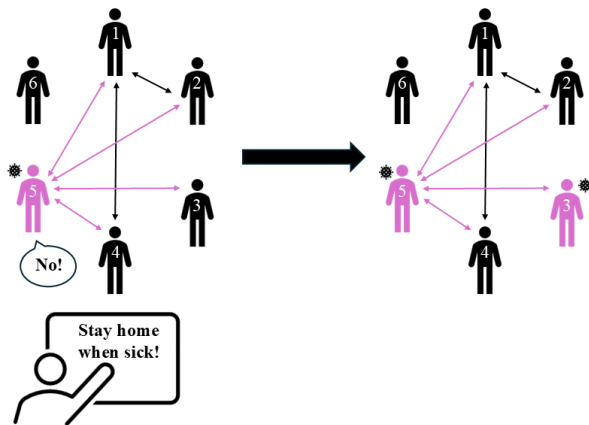
Example III: Intervention Affects Transmission, Not Network



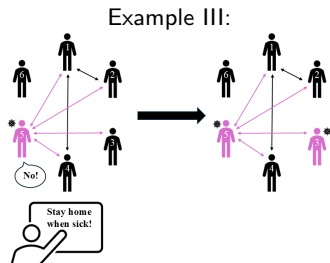
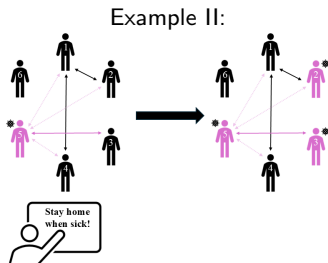
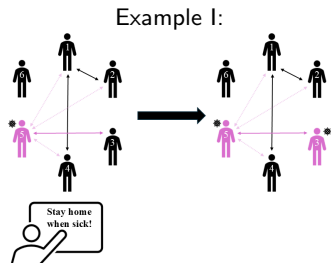
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Central Question: Does the Intervention Affect the Network?



In Examples I and III, the answer is yes

eX-FLU Potential Outcomes

Potential outcomes are defined for both the networks, and ILI infections:

$$\overline{\mathbf{A}}(\mathbf{z}), \quad \overline{\mathbf{Y}}(\mathbf{z}) \quad \text{for } \mathbf{z} \in \mathcal{Z}$$

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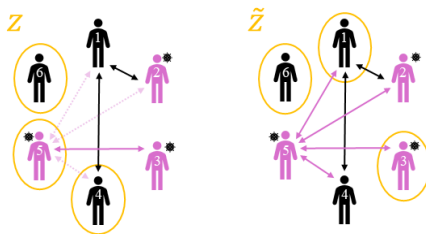
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- Assume **causal consistency**:

$$\overline{\mathbf{A}} = \overline{\mathbf{A}}(\mathbf{Z}) \text{ and } \overline{\mathbf{Y}} = \overline{\mathbf{Y}}(\mathbf{Z})$$

eX-FLU Null Hypotheses

$$H_0^Y : \overline{Y}(\mathbf{z}) = \overline{Y}(\tilde{\mathbf{z}}) \text{ for all } \mathbf{z}, \tilde{\mathbf{z}} \in \mathcal{Z}$$

(“the intervention has no effect on the infections”)



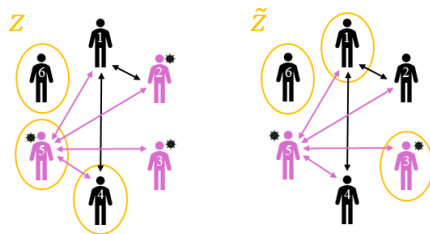
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$$H_0^Y : \bar{Y}(z) = \bar{Y}(\tilde{z}) \text{ for all } z, \tilde{z} \in \mathcal{Z}$$

(“the intervention has no effect on the infections”)

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(“the intervention has no effect the networks”)



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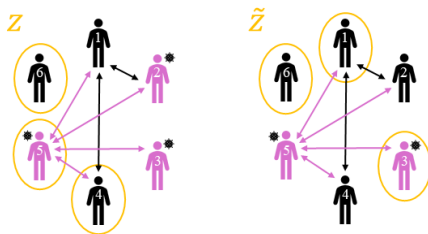
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$$H_0^\sharp = H_0^Y \cap H_0^A : \bar{Y}(z) = \bar{Y}(\tilde{z}) \text{ and } \bar{A}(z) = \bar{A}(\tilde{z}) \text{ for all } z, \tilde{z} \in \mathcal{Z}$$

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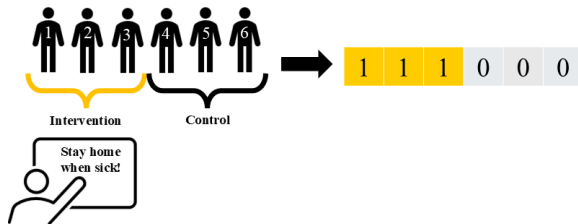
Previous analyses of the eX-FLU trial ([Alexandria et al., 2023](#)) tested $H_0^\#$, but no analyses have tested H_0^Y

eX-FLU Observed Data

- Baseline randomization assignments

$\mathbf{Z} = (Z_1, \dots, Z_n) \in \mathcal{Z}$, for

$Z_i = \mathbb{1}(\text{student } i \text{ gets intervention})$



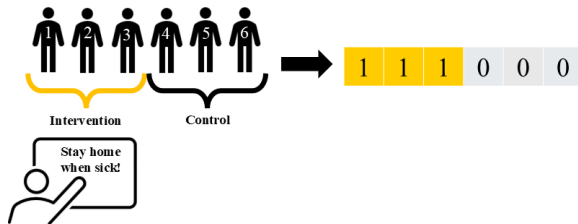
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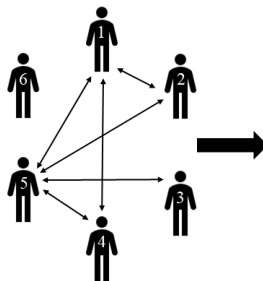
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 $A_{ij}^t = \mathbb{1}(\text{students } i, j \text{ in contact at week } t)$



	1	2	3	4	5	6
1	0	1	0	1	1	0
2	1	0	0	0	1	0
3	0	0	0	0	1	0
4	1	0	0	0	1	0
5	1	1	1	1	0	0
6	0	0	0	0	0	0

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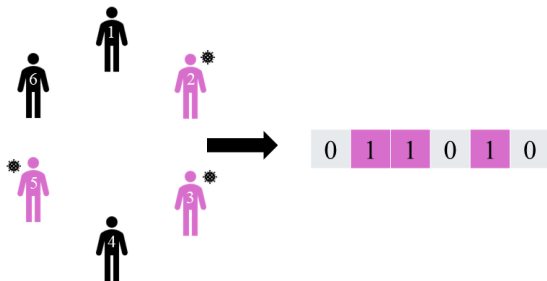
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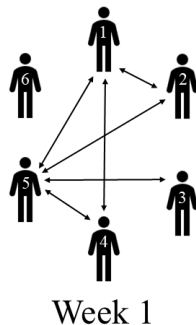
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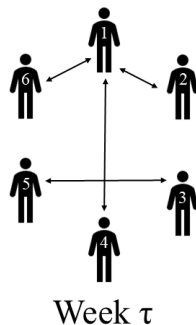


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- network history: $\bar{\mathbf{A}} = \{\mathbf{A}^t\}_{t=1}^\tau$

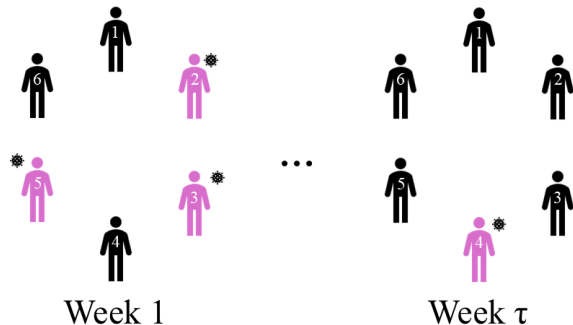


...

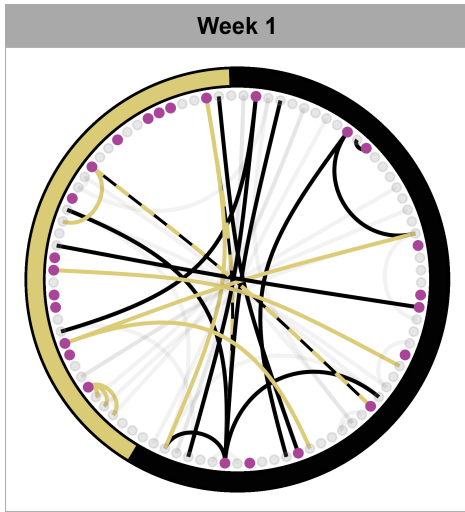


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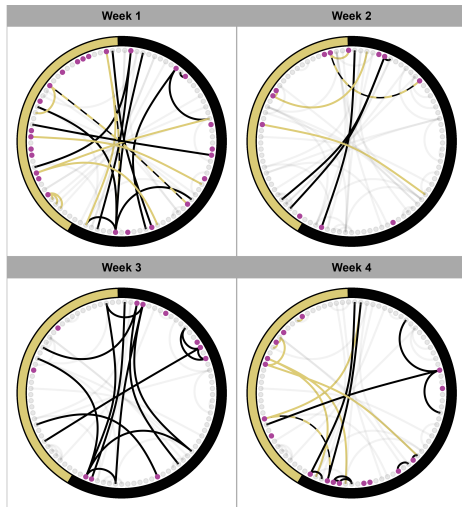


eX-FLU: Descriptive Results



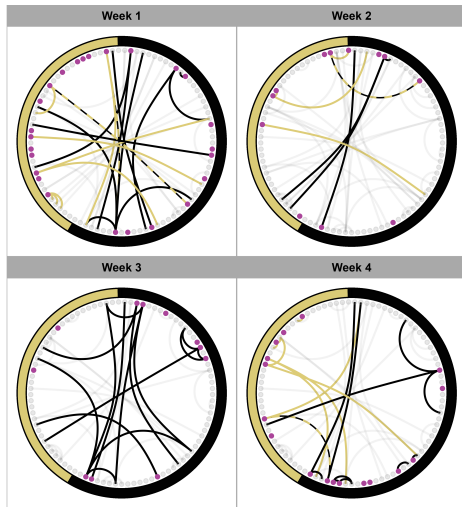
- 93 (out of 579) students with at least one infection
- Nodes represent students
 - ▶ gray → uninfected
 - ▶ pink → infected
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 - ▶ black → control
 - ▶ yellow → encouraged to isolate

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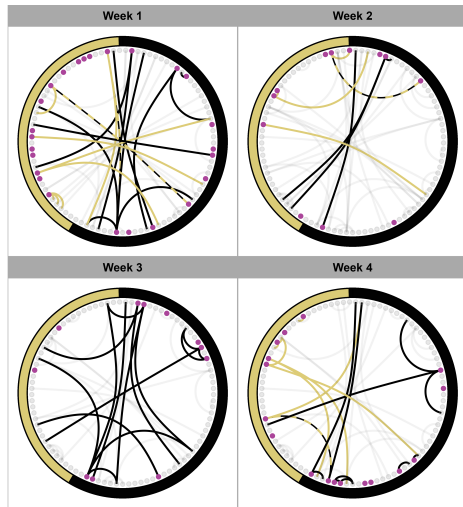
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- Outer ring represents intervention assignment
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- Bold edges represent possible transmission events
 - ▶ Proportion of possible transmission events attributable to students in the intervention group:

$$\begin{aligned}
 T &= \frac{\text{number of yellow edges}}{\text{total number of edges}} = \\
 &= \frac{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} Z_i Y_i^{k-1} A_{ij}^{k-1}}{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} Y_i^{k-1} A_{ij}^{k-1}} \\
 &= 0.359
 \end{aligned}$$

Back to the Central Question

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- Is this due to random chance?
- Or is this because the intervention has some affect on the networks or on transmission?

Testing $H_0^\#$

“How unlikely are the observed data under $H_0^\#$?”

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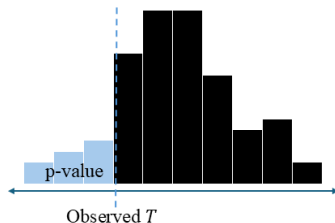
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$$F_T(t|H_0^\sharp) = \sum_{\mathbf{z} \in \mathcal{Z}} \Pr(\mathbf{Z} = \mathbf{z}) \mathbb{1}\{T(\mathbf{z}, \bar{\mathbf{A}}, \bar{\mathbf{Y}}) \leq t\}$$



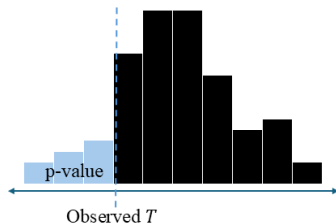
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- Reject $H_0^\sharp$ if $\rho^\sharp \leq 0.05$



Testing $H_0^\#$

- This test of $H_0^\#$ controls the **type I error rate** exactly:

$$\Pr(\rho^\# \leq \alpha | H_0^\#) \leq \alpha$$

for any $\alpha \in [0, 1]$, for any sample size, and for any choice of test statistic

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- ▶ In general, T should be chosen such that small values indicate departures from $H_0^\#$

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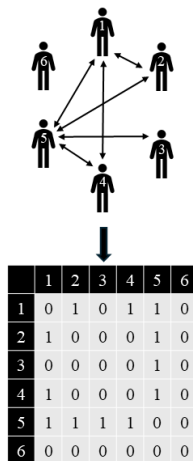
Question: How to model $q(\bar{\mathbf{a}}, \mathbf{z}; \theta)$?

Exponential Family Random Graph Models

Goal: model a network ($\mathbf{A}$) given covariates ($\mathbf{X}$)

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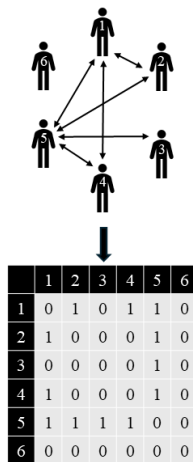


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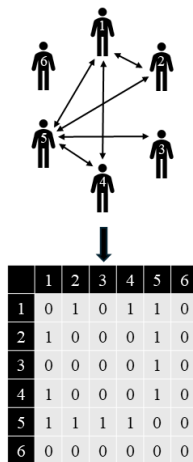
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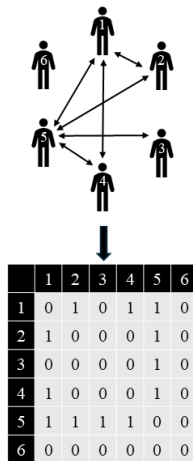
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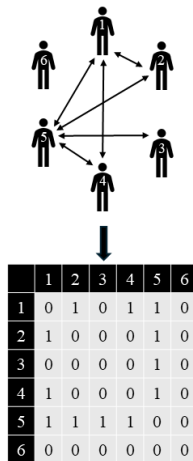
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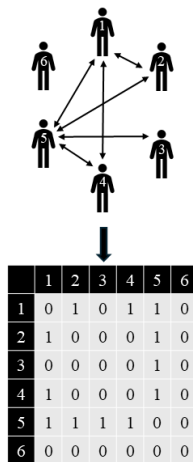
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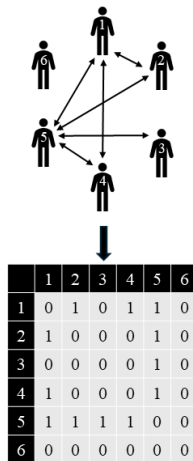
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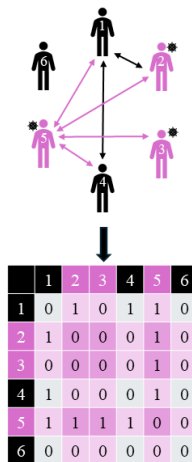
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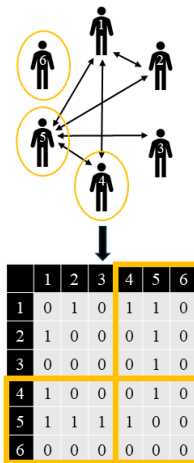
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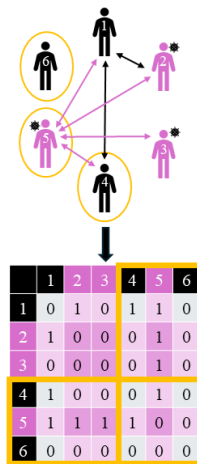
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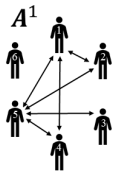
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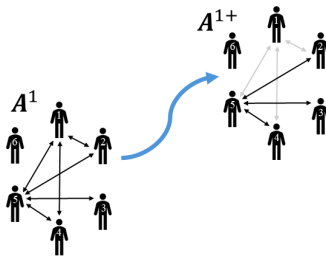
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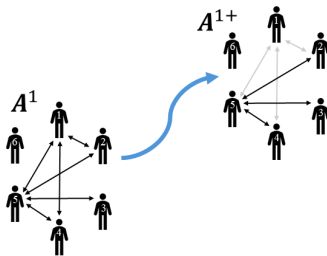


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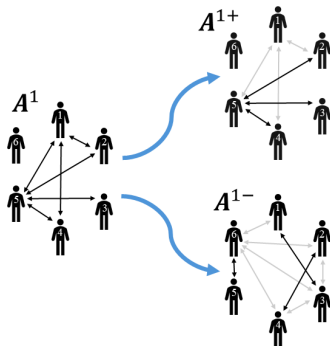
$$\Pr_{\theta^+}(\mathbf{A}^{k+1} = \mathbf{a}^{k+1} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^+ \cdot \mathbf{g}^+(\mathbf{a}^{k+1}, \mathbf{x})\}}{\kappa\{\theta^+, \mathcal{A}^+(\mathbf{a}^k), \mathbf{x}\}}$$



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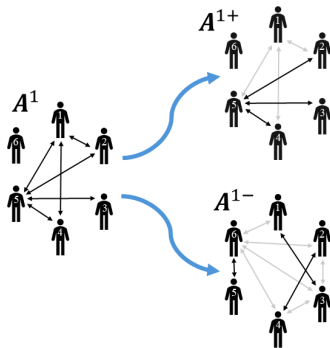


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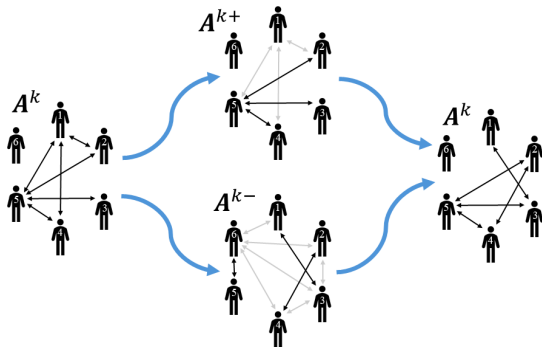
$$\Pr(\mathbf{A}^{k-} = \mathbf{a}^{k-} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^- \cdot \mathbf{g}^-(\mathbf{a}^{k-}, \mathbf{x})\}}{\kappa\{\theta^-, \mathcal{A}^-(\mathbf{a}^k), \mathbf{x}\}}.$$



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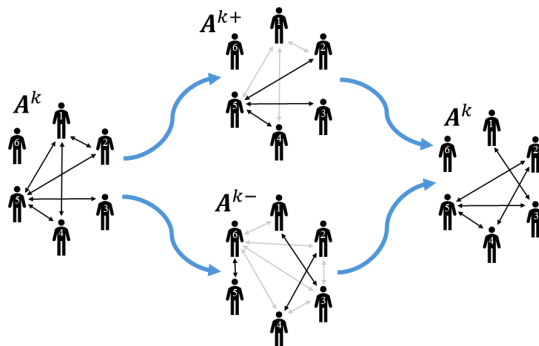


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$$\mathbf{A}^{k+1} = \underbrace{\mathbf{A}^k}_{\text{previous network}} \cup \underbrace{(\mathbf{A}^{k+} - \mathbf{A}^k)}_{\text{new edges formed}} - \underbrace{(\mathbf{A}^k - \mathbf{A}^{k-})}_{\text{old edges not persisting}}$$

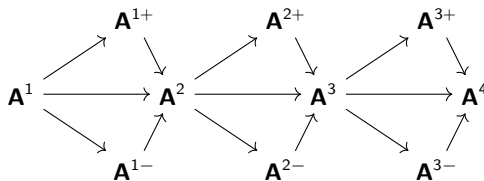


Separability of STERGMs

- 1 $\mathbf{A}^{k+} \perp\!\!\!\perp \mathbf{A}^{k-} | \mathbf{A}^k$, i.e., the formation and persistence processes are conditionally independent given the network at time k

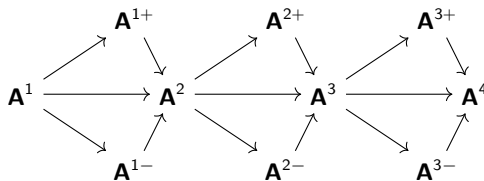
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- 2 the parameter space for $\theta = (\theta^+, \theta^-)$ is the product of the parameter spaces for θ^+ and θ^-



Simulation Study

- $n \in \{24, 48\}$ students
- $\tau \in \{5, 10\}$ weeks
- 50:50 cluster randomization with 12 equal-sized clusters

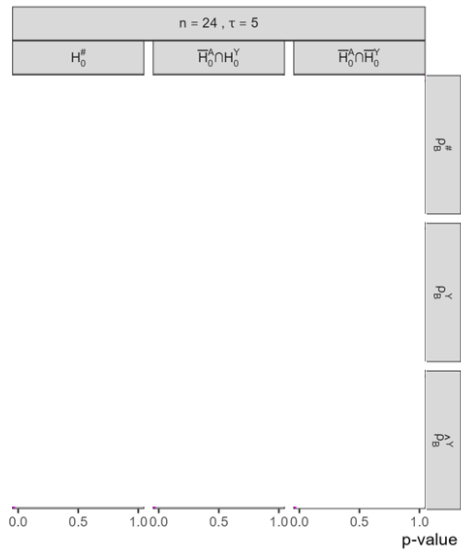
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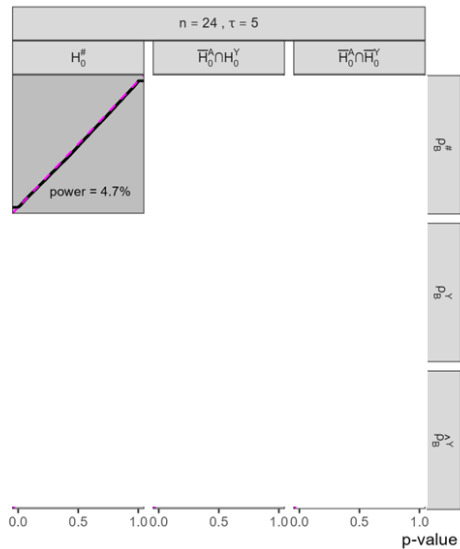
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 - 2 ρ_B^Y : testing H_0^Y using known q
 - 3 $\hat{\rho}_B^Y$: testing H_0^Y using estimated q

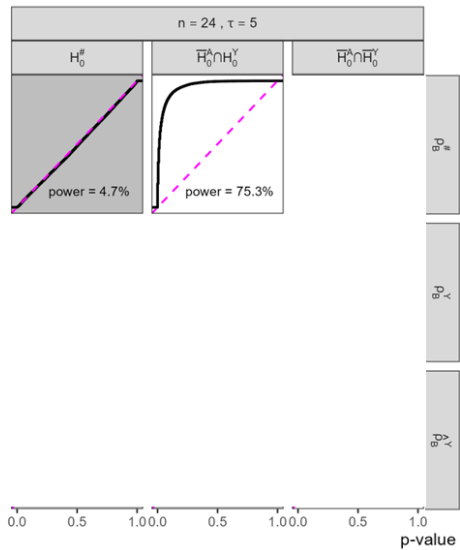
Simulation Results



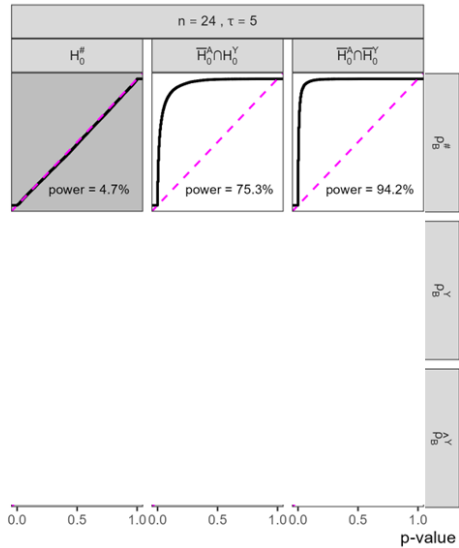
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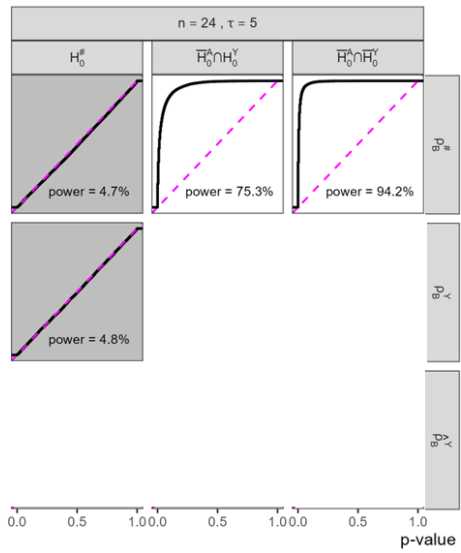
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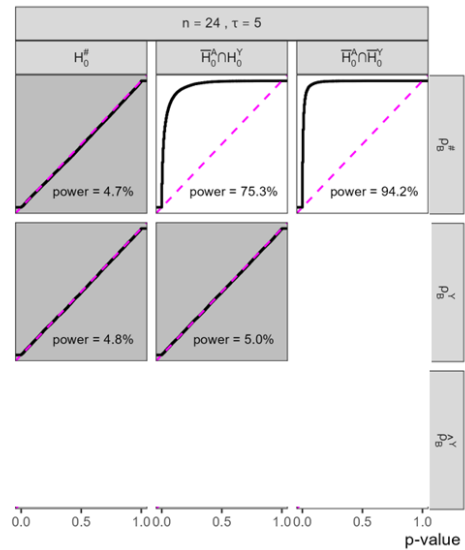
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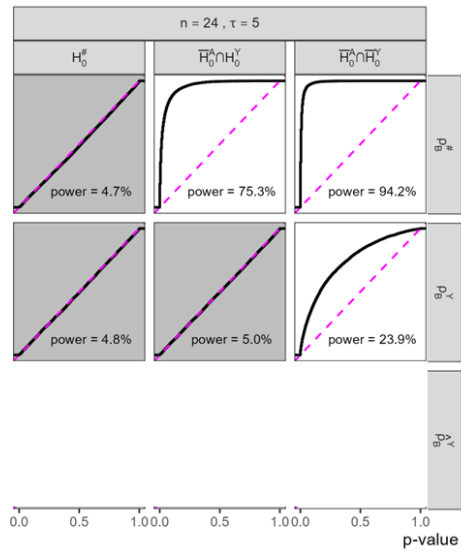
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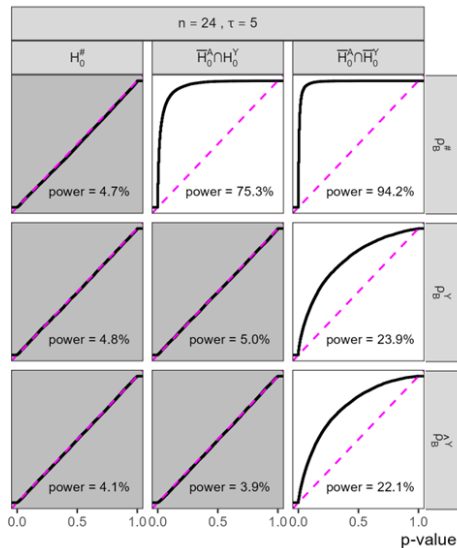
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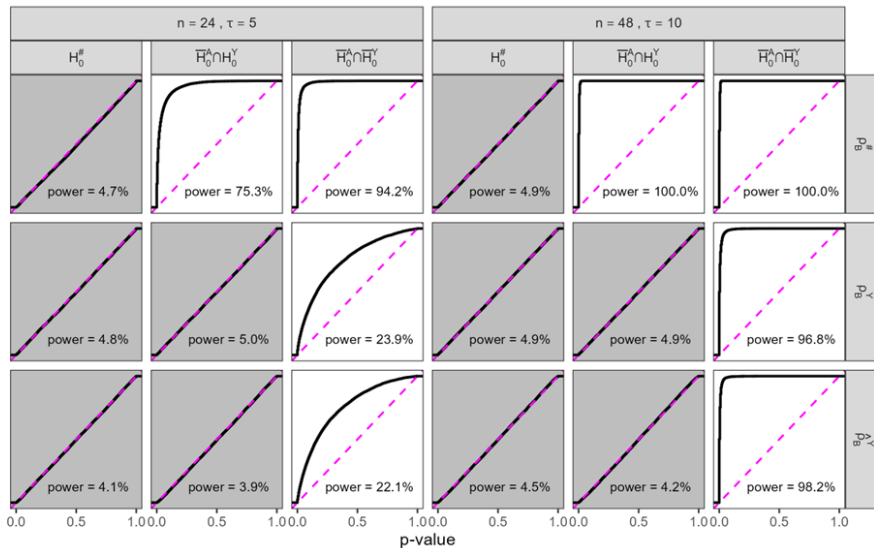
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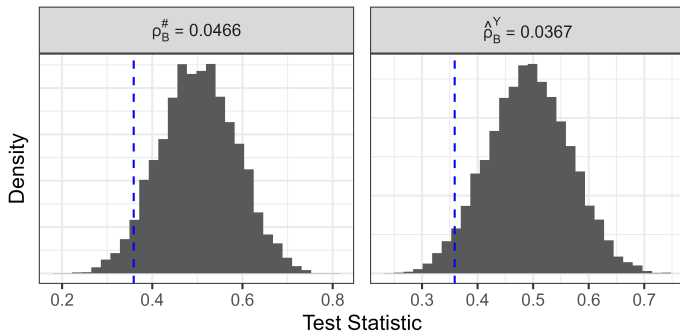
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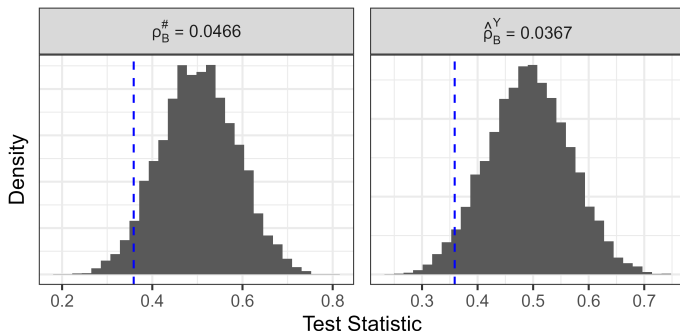


eX-FLU: Hypothesis Test Results



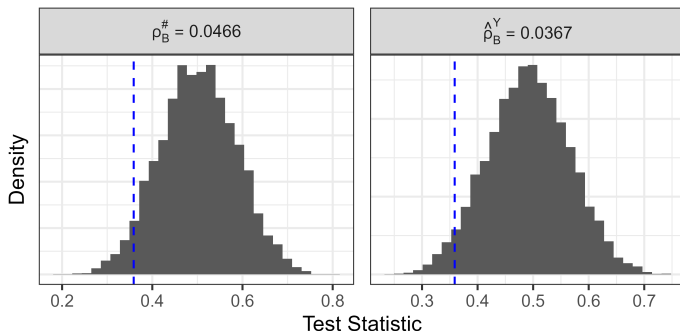
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- Encouragement to isolate specifically affects influenza-like illness ($\hat{\rho}_B^Y = 0.0367$)

Summary of Project 2

- Developed a new randomization-based inference procedure
- Established theoretical properties (asymptotic control of Type I error)
- Demonstrated empirical performance through simulations
- Applied the method to assess a protective effect of an encouragement-to-isolate intervention on a college campus
- Status: preparing submission to *PNAS* (target: December)

Project 3 : Causal Inference from Cluster Randomized Trials with Differential Nonresponse

Brian Richardson, Bonnie Shook-Sa, Michael Hudgens,

PopART Trial

- **PopART**: large **cluster randomized trial** designed to evaluate a combination **HIV prevention intervention** in Zambia and South Africa ([Hayes et al., 2014](#))

Hayes et al. *Trials* 2014, **15**:57
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STUDY PROTOCOL

Open Access

HPTN 071 (PopART): Rationale and design of a cluster-randomised trial of the population impact of an HIV combination prevention intervention including universal testing and treatment – a study protocol for a cluster randomised trial

Richard Hayes¹, Helen Ayles^{2,3}, Nulda Beyers⁴, Kalpana Sabapathy^{1*}, Sian Floyd¹, Kwame Shanaube³, Peter Bock⁴, Sam Griffith⁵, Ayana Moore⁵, Deborah Watson-Jones², Christophe Fraser⁶, Sten H Vermund⁷, Sarah Fidler⁸ and The HPTN 071 (PopART) Study Team

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 - ▶ Arm A: annual home-based HIV testing, promotion of medical male circumcision for HIV-negative men, and offer of immediate ART for those testing HIV-positive
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- **Outcome**: HIV Incidence

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PopART Trial

- **Initial Findings:** no protective effect of the Arm A intervention ([Hayes et al., 2019](#))
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Effect of Universal Testing and Treatment on HIV Incidence — HPTN 071 (PopART)

R.J. Hayes, D. Donnell, S. Floyd, N. Mandla, J. Bwalya, K. Sabapathy, B. Yang, M. Phiri, A. Schaap, S.H. Eshleman, E. Piwowar-Manning, B. Kosloff, A. James, T. Skalland, E. Wilson, L. Emel, D. Macleod, R. Dunbar, M. Simwinga, N. Makola, V. Bond, G. Hodidinott, A. Moore, S. Griffith, N. Deshmane Sista, S.H. Vermund, W. El-Sadr, D.N. Burns, J.R. Hargreaves, K. Hauck, C. Fraser, K. Shanaube, P. Bock, N. Beyers, H. Ayles, and S. Fidler, for the HPTN 071 (PopART) Study Team

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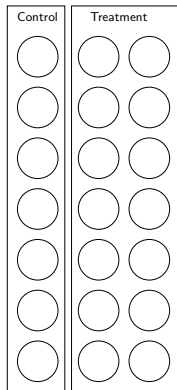
- **Initial Findings:** no protective effect of the Arm A intervention ([Hayes et al., 2019](#))
 - ▶ “unanticipated and not consistent with the data on viral suppression”
- **Later Analyses:** identified possibly differential nonresponse that could bias trial results ([Shook-Sa et al., 2025+](#))
 - ▶ Men were less likely to respond (i.e., have data collected)
 - ▶ The intervention appeared more effective among men



Effect of Universal Testing and Treatment on HIV Incidence — HPTN 071 (PopART)

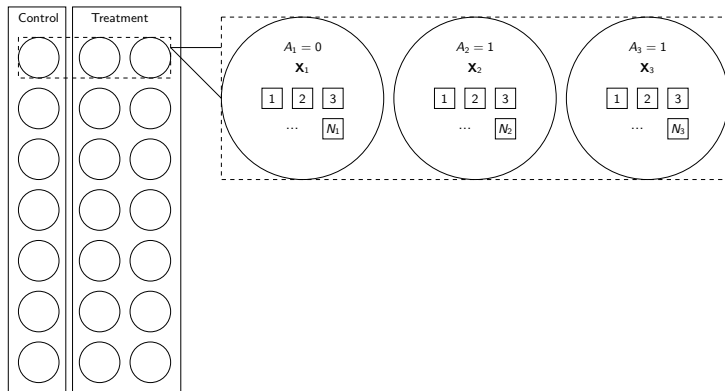
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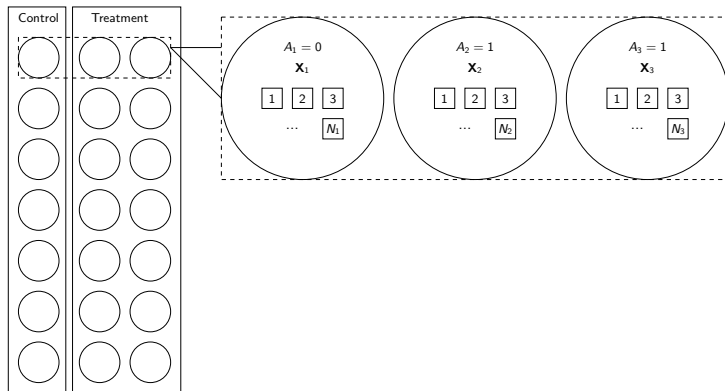
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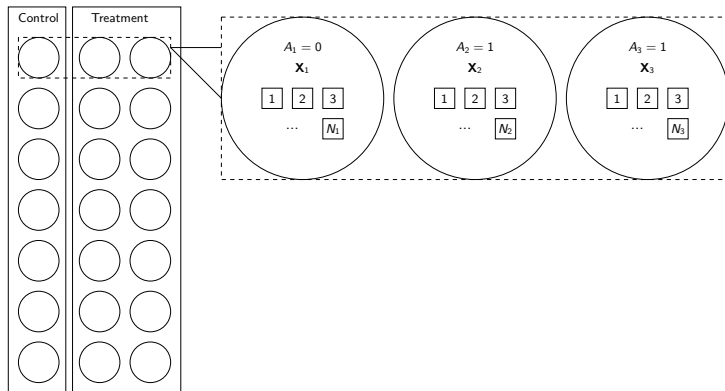
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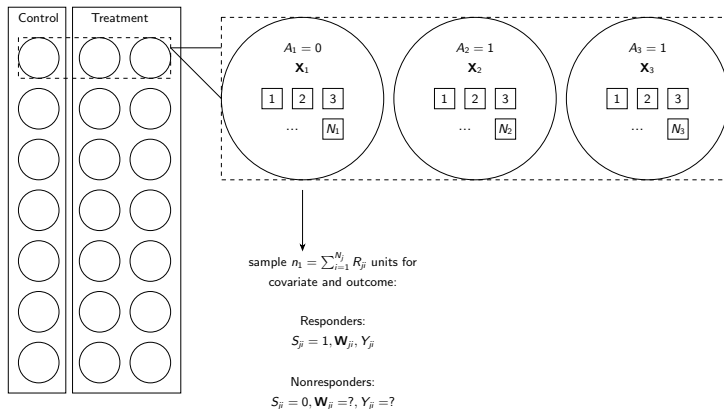
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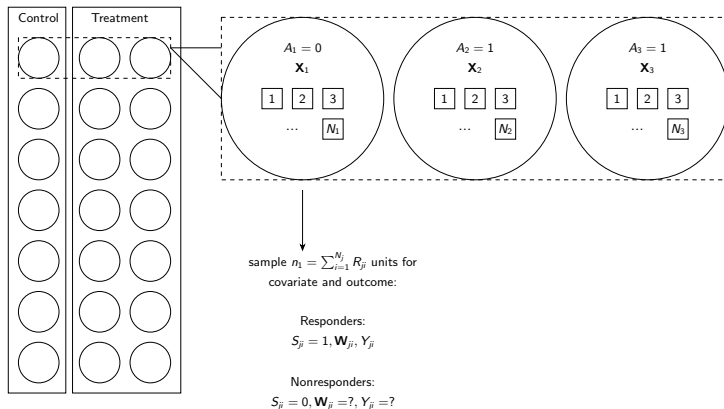
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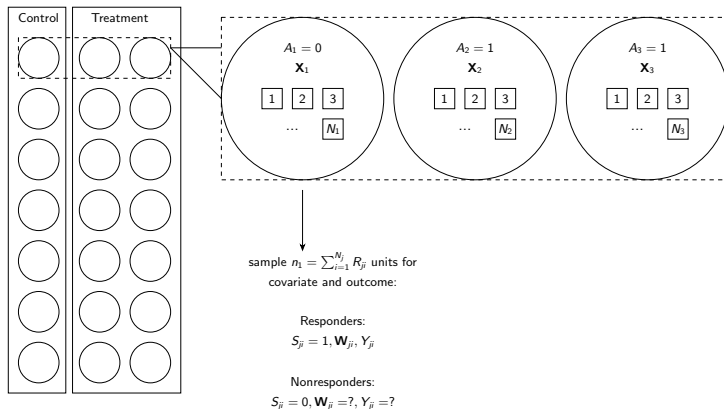
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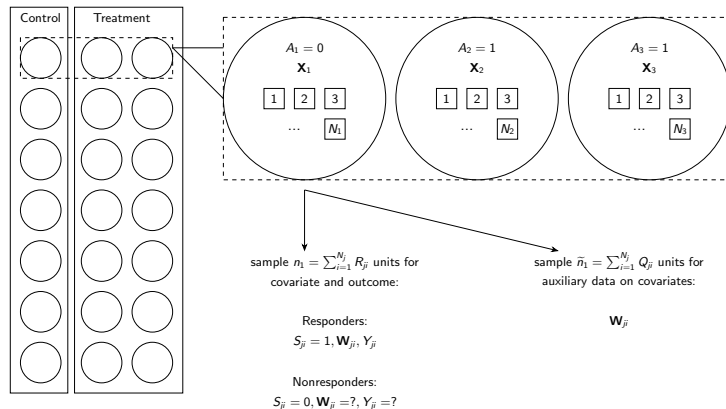
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- **individual-level potential outcomes:**

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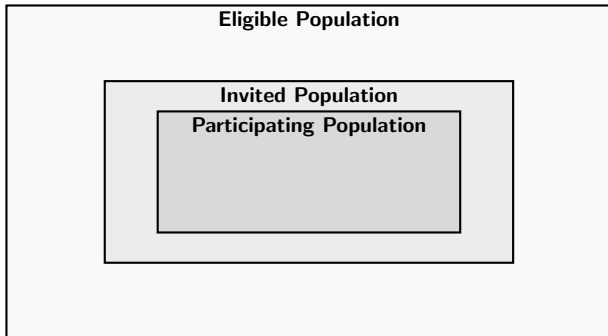
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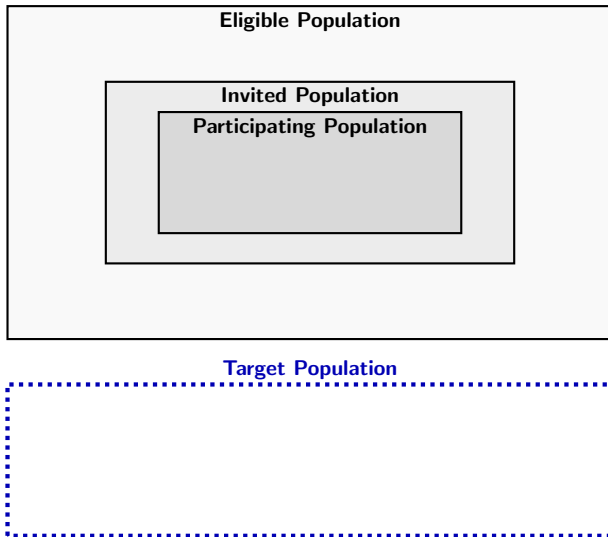
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Extending Inference from Randomized Controlled Trials

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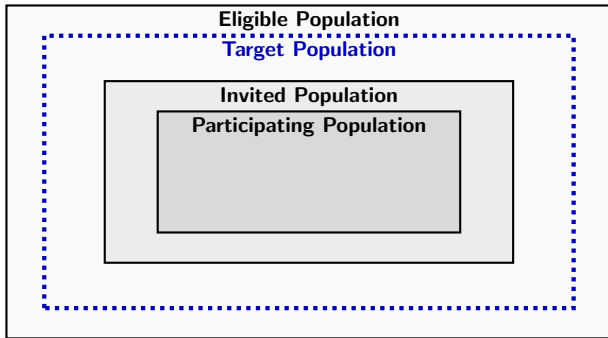


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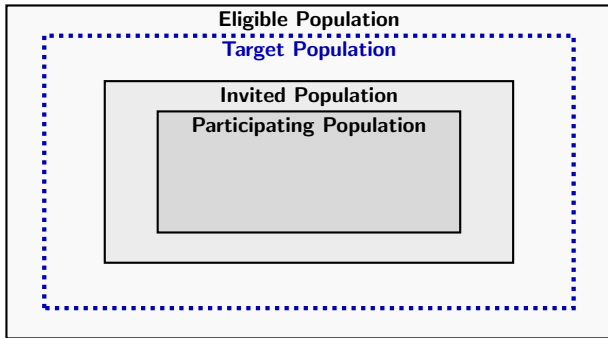
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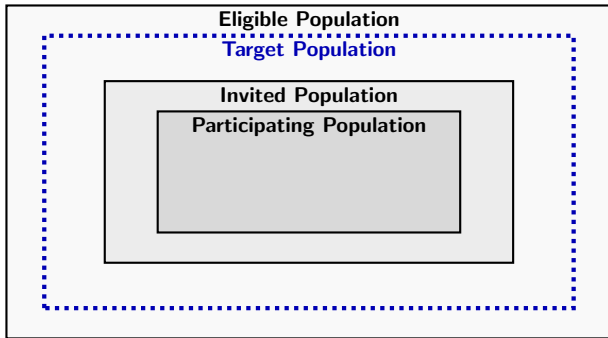
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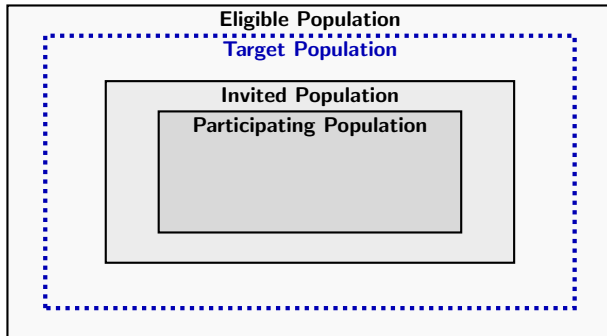
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- **Goal:** generalize results to trial-eligible population using data from responders

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Corrected Score Functions: What?

- Suppose the oracle score function is **conditionally unbiased**, meaning

$$E\{\Psi_0(Y, \mathbf{L}, \mathbf{A}; \theta) | \mathbf{A}\} = \mathbf{0}.$$

- Define the corrected score function as

$$\Psi_{CS}(Y, \mathbf{L}, \mathbf{A}^*; \theta) = E \left[\text{Re} \left\{ \Psi_0(Y, \mathbf{L}, \tilde{\mathbf{A}}; \theta) \right\} \mid Y, \mathbf{L}, \mathbf{A}^* \right],$$

where $\tilde{\mathbf{A}} = \mathbf{A}^* + i\tilde{\epsilon}$, $i = \sqrt{-1}$, $\text{Re}(\cdot)$ denotes the real component of a complex number, and $\tilde{\epsilon} \sim \mathcal{N}(\mathbf{0}, \Sigma_{me})$.

- Then

$$\begin{aligned} E\{\Psi_{CS}(Y, \mathbf{L}, \mathbf{A}^*; \theta) \mid Y, \mathbf{L}, \mathbf{A}\} &= \Psi_0(Y, \mathbf{L}, \mathbf{A}; \theta) \\ \implies E[E\{\Psi_{CS}(Y, \mathbf{L}, \mathbf{A}^*; \theta) \mid Y, \mathbf{L}, \mathbf{A}\}] &= E\{\Psi_0(Y, \mathbf{L}, \mathbf{A}; \theta)\} \\ \implies E\{\Psi_{CS}(Y, \mathbf{L}, \mathbf{A}^*; \theta)\} &= \mathbf{0} \end{aligned}$$

Corrected Score Functions: Why?

The key result ([Novick and Stefanski, 2002](#)) for corrected score functions is that, for a smooth enough function $\mathbf{f}$, the function $\tilde{\mathbf{f}}$ defined by

$$\tilde{\mathbf{f}}(\mathbf{A}^*) \equiv \mathbb{E}[\text{Re}\{\mathbf{f}(\mathbf{A}^* + i\tilde{\epsilon})\} | \mathbf{A}, \mathbf{A}^*]$$

does not depend on $\mathbf{A}$ and satisfies

$$\mathbb{E}\{\tilde{\mathbf{f}}(\mathbf{A}^*) | \mathbf{A}\} = \mathbf{f}(\mathbf{A}).$$

The proof of this for the special case $f(\mathbf{A}) = \exp(\mathbf{c}\mathbf{A}^T)$ is illustrative.

Corrected Score Functions: Why?

CLAIM: $E\{\tilde{f}(\mathbf{A}^*)|\mathbf{A}\} = f(\mathbf{A})$ for $f(\mathbf{A}) = \exp(\mathbf{c}\mathbf{A}^T)$.

Corrected Score Functions: Why?

CLAIM: $E\{\tilde{f}(\mathbf{A}^*)|\mathbf{A}\} = f(\mathbf{A})$ for $f(\mathbf{A}) = \exp(\mathbf{c}\mathbf{A}^T)$.

$$\begin{aligned}\tilde{f}(\mathbf{A}^*) &\equiv E(\text{Re}[\exp\{\mathbf{c}(\mathbf{A}^* + i\tilde{\epsilon})^T\}]|\mathbf{A}, \mathbf{A}^*) \\ &= \exp(\mathbf{c}\mathbf{A}^{*T}) \underbrace{\text{Re}[E\{\exp(i\mathbf{c}\tilde{\epsilon}^T)\}]}_{\text{normal c.f.}} = \exp(\mathbf{c}\mathbf{A}^{*T}) \exp\left(-\frac{1}{2}\mathbf{c}\boldsymbol{\Sigma}_{me}\mathbf{c}^T\right)\end{aligned}$$

Corrected Score Functions: Why?

CLAIM: $E\{\tilde{f}(\mathbf{A}^*)|\mathbf{A}\} = f(\mathbf{A})$ for $f(\mathbf{A}) = \exp(\mathbf{c}\mathbf{A}^T)$.

$$\begin{aligned}\tilde{f}(\mathbf{A}^*) &\equiv E(\text{Re}[\exp\{\mathbf{c}(\mathbf{A}^* + i\tilde{\epsilon})^T\}]|\mathbf{A}, \mathbf{A}^*) \\ &= \exp(\mathbf{c}\mathbf{A}^{*T}) \underbrace{\text{Re}[E\{\exp(i\mathbf{c}\tilde{\epsilon}^T)\}]}_{\text{normal c.f.}} = \exp(\mathbf{c}\mathbf{A}^{*T}) \exp\left(-\frac{1}{2}\mathbf{c}\boldsymbol{\Sigma}_{me}\mathbf{c}^T\right) \\ \Rightarrow E\{\tilde{f}(\mathbf{A}^*)|\mathbf{A}\} &= E\{\exp(\mathbf{c}\mathbf{A}^{*T})|\mathbf{A}\} \exp\left(-\frac{1}{2}\mathbf{c}\boldsymbol{\Sigma}_{me}\mathbf{c}^T\right) \\ &= E\{\exp\{\mathbf{c}(\mathbf{A}^T + \epsilon^T)\}|\mathbf{A}\} \exp\left(-\frac{1}{2}\mathbf{c}\boldsymbol{\Sigma}_{me}\mathbf{c}^T\right) \\ &= \exp(\mathbf{c}\mathbf{A}^T) \underbrace{E\{\exp(\mathbf{c}\epsilon^T)\}}_{\text{normal m.g.f.}} \exp\left(-\frac{1}{2}\mathbf{c}\boldsymbol{\Sigma}_{me}\mathbf{c}^T\right) \\ &= \exp(\mathbf{c}\mathbf{A}^T) \exp\left(\frac{1}{2}\mathbf{c}\boldsymbol{\Sigma}_{me}\mathbf{c}^T\right) \exp\left(-\frac{1}{2}\mathbf{c}\boldsymbol{\Sigma}_{me}\mathbf{c}^T\right) \\ &= \exp(\mathbf{c}\mathbf{A}^T) = f(\mathbf{A})\end{aligned}$$

Additional Simulation Results: G-Formula

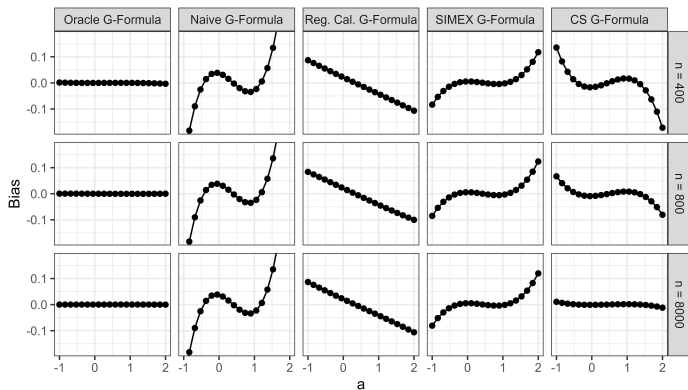


Figure: Estimated dose-response curve bias for the oracle, naive, regression calibration, SIMEX, and CS g-formula estimators. Bias refers to the average bias across 1000 simulated data sets for each method evaluated at each point on the horizontal axis, corresponding to setting the true exposure to $a \in [-1, 2]$. For the naive estimator, biases outside of $[-0.18, 0.18]$ are excluded from the plot.

Additional Simulation Results: IPW

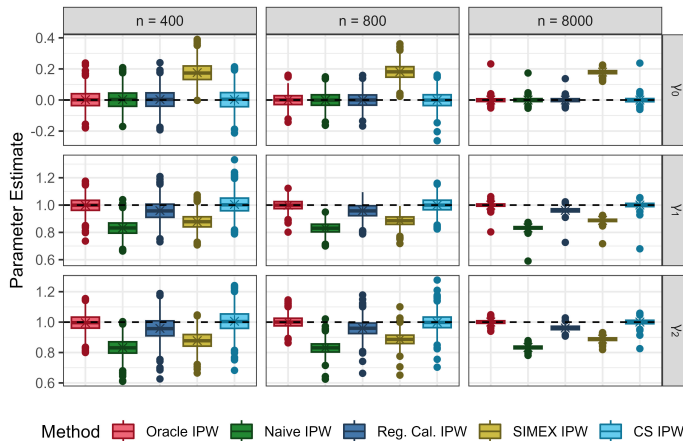


Figure: Empirical distribution of MSM parameter estimates from the simulation study of IPW estimators, using the oracle, naive, regression calibration (reg. cal.), simulation-extrapolation (SIMEX), and corrected score (CS) IPW estimators.

Additional Simulation Results: Doubly Robust Estimator

Table: Results from the simulation study of DR estimators. UC: uncorrected empirical sandwich variance estimator, BC: bias-corrected empirical sandwich variance estimator. PS indicates the propensity score model is correctly specified; OR indicates the outcome regression is correctly specified.

n	Correct Specifications	Method	Bias	ESE	UC		BC	
					ASE	Cov	ASE	Cov
2000	PS and OR	CS DR	0.2	8.5	7.9	92.6	7.9	92.9
		CS G-Formula	0.0	6.8	6.6	94.2	6.6	94.3
		CS IPW	0.1	8.9	8.3	93.5	8.4	94.0
	PS Only	CS DR	-0.1	8.6	8.0	93.3	8.1	93.4
		CS G-Formula	-14.9	6.7	6.6	39.7	6.6	40.0
		CS IPW	0.1	8.9	8.3	93.5	8.4	94.0
	OR Only	CS DR	0.1	7.7	7.2	93.0	7.2	93.0
		CS G-Formula	0.0	6.8	6.6	94.2	6.6	94.3
		CS IPW	-14.7	7.6	7.3	47.9	7.3	47.9
	Neither	CS DR	-14.7	7.5	7.1	47.0	7.2	47.3
		CS G-Formula	-14.9	6.7	6.6	39.7	6.6	40.0
		CS IPW	-14.7	7.6	7.3	47.9	7.3	47.9

Additional Simulation Results: Estimated Measurement Error Variance

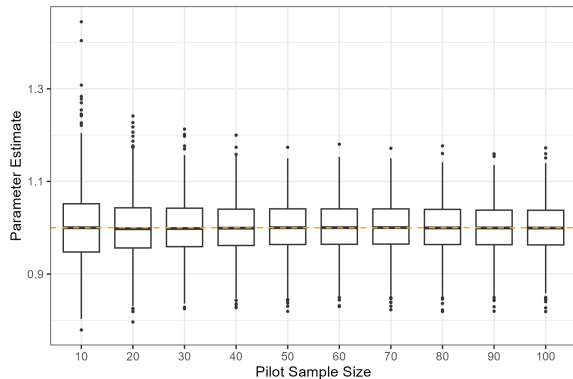


Figure: Empirical distribution of estimated γ_1 versus pilot study sample size n_p , using the corrected score (CS) IPW estimator with estimated measurement error covariance.

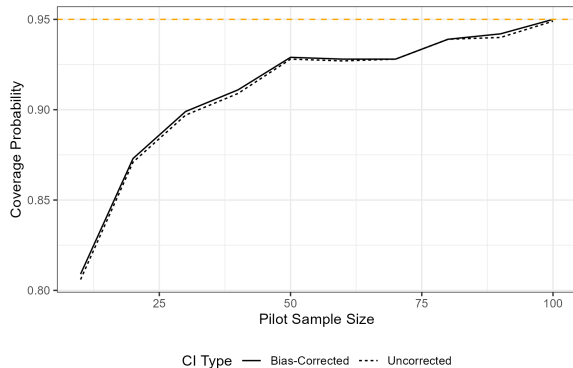


Figure: Empirical coverage probabilities of 95% confidence intervals for γ_1 versus pilot study sample size n_p , using the corrected score (CS) IPW estimator with estimated measurement error covariance.

Additional Simulation Results: Near-Positivity Violation

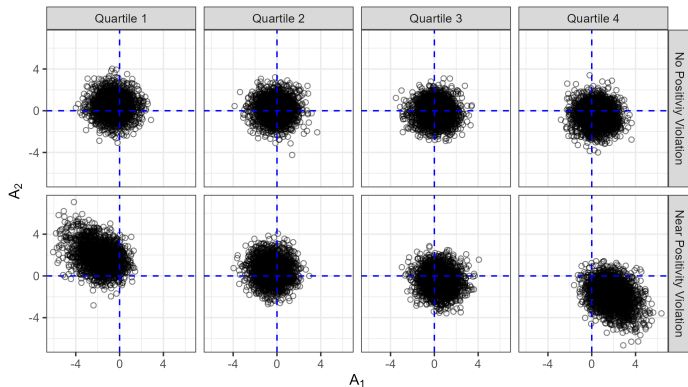


Figure: True exposure values (A_1, A_2) by quartile of L from one simulated data set of sample size $n = 8000$, with and without a near positivity violation.

Additional Simulation Results: Near-Positivity Violation

Table: Results from the simulation study with a near positivity violation. UC: uncorrected empirical sandwich variance estimator, BC: bias-corrected empirical sandwich variance estimator, n: sample size; Bias: 100 times the average bias across simulated data sets for each method; ESE: 100 times the standard deviation of parameter estimates; ASE: 100 times the average of estimated standard errors; Cov: Empirical percent coverage of 95% confidence intervals for each method.

n	Method	Parameter	Bias	ESE	UC		BC	
					ASE	Cov	ASE	Cov
8000	Oracle IPW	γ_0	-0.8	10.8	6.4	90.4	6.6	91.0
		γ_1	4.2	8.2	4.5	65.6	5.2	69.9
		γ_2	-3.6	8.2	4.7	64.3	5.4	68.7
	Naive IPW	γ_0	-0.4	10.7	6.8	89.1	7.0	89.5
		γ_1	-13.0	7.9	4.6	17.9	5.1	21.6
		γ_2	-19.9	8.0	4.6	10.2	5.1	12.2
	CS IPW	γ_0	-2.3	46.0	44.9	89.9	74.3	90.1
		γ_1	4.4	66.4	61.9	68.0	106.0	71.5
		γ_2	-4.3	33.0	31.0	76.6	49.8	78.8

Additional Simulation Results: Multiplicative Measurement Error

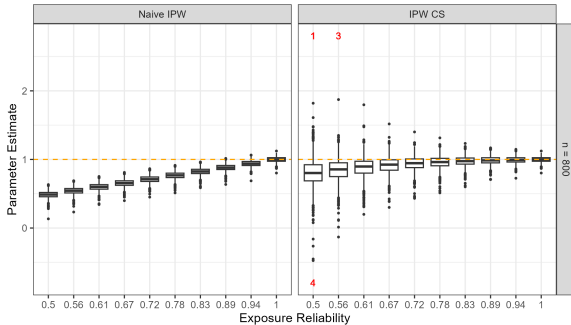


Figure: Empirical distribution of estimated γ_1 versus exposure reliability $\text{Var}(A_1)/\text{Var}(A_1^*) = \text{Var}(A_2)/\text{Var}(A_2^*)$, using naive and corrected score (CS) IPW estimators and for a multiplicative measurement error data generating process. For the CS IPW estimator, the number values beyond the ranges of the plot are shown in red on the tails of the boxplots.

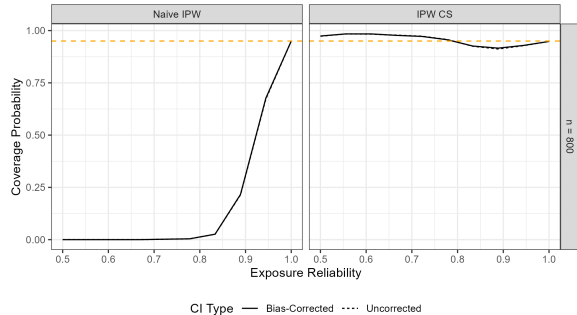


Figure: Empirical coverage probabilities of 95% confidence intervals for γ_1 versus exposure reliability $\text{Var}(A_1)/\text{Var}(A_1^*) = \text{Var}(A_2)/\text{Var}(A_2^*)$, using naive and corrected score (CS) IPW estimators for a multiplicative measurement error data generating process.

Choice of Test Statistic

Test statistics: “proportion of possible transmission events attributable to students in the intervention group:”

$$T^{**}(\mathbf{Z}, \bar{\mathbf{A}}, \bar{\mathbf{Y}}) = \frac{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} Z_i E_{ijk}^{**}}{\sum_{k=2}^{\tau} \sum_{i=1}^n \sum_{j>i} E_{ijk}^{**}}.$$

	From Infected to Infected	From Infected
Contact at $k - 1$	$E_{ijk}^{11} = Y_i^{k-1} A_{ij}^{k-1} Y_j^k$	$E_{ijk}^{12} = Y_i^{k-1} A_{ij}^{k-1}$
Contact at k	$E_{ijk}^{21} = Y_i^{k-1} A_{ij}^k Y_j^k$	$E_{ijk}^{22} = Y_i^{k-1} A_{ij}^k$
Contact at $k - 1$ or k	$E_{ijk}^{31} = Y_i^{k-1} (A_{ij}^{k-1} \vee A_{ij}^k) Y_j^k$	$E_{ijk}^{32} = Y_i^{k-1} (A_{ij}^{k-1} \vee A_{ij}^k)$
Contact at $k - 1$ and k	$E_{ijk}^{41} = Y_i^{k-1} (A_{ij}^{k-1} * A_{ij}^k) Y_j^k$	$E_{ijk}^{42} = Y_i^{k-1} (A_{ij}^{k-1} * A_{ij}^k)$

Table: Definitions of a possible transmission event E_{ijk} from student i to student j at time k . Y_i^k is an indicator for student i being infected at week k , A_{ij}^k is an indicator for students i and j being in contact at week k , and $a \vee b$ denotes the maximum of a and b .

ERGM Model Formulation

$$\Pr_{\theta}(\mathbf{A} = \mathbf{a} | \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}}{\kappa(\theta, \mathcal{A}, \mathbf{x})},$$

$$\kappa(\theta, \mathcal{A}, \mathbf{x}) = \sum_{\mathbf{a} \in \mathcal{A}} \exp\{\theta \cdot \mathbf{g}(\mathbf{a}, \mathbf{x})\}$$

For example, in the simulation study and eX-FLU application,

$$\mathbf{g}(\mathbf{a}, \mathbf{x}) = \begin{bmatrix} \# \text{ edges} \\ \# \text{ edges touching a treated node} \\ \# \text{ edges touching an infected node} \\ \# \text{ edges touching a treated and infected node} \\ \# \text{ edges between roommate pairs} \end{bmatrix} = \begin{bmatrix} \sum_{i,j} a_{ij} \\ \sum_{i,j} a_{ij}(z_i + z_j) \\ \sum_{i,j} a_{ij}(y_i + y_j) \\ \sum_{i,j} a_{ij}(z_i y_i + z_j y_j) \\ \sum_{i,j} a_{ij} \mathbb{1}(i, j \text{ roommates}) \end{bmatrix}$$

ERGM Change Statistic Model Formulation

- **Change statistic:** $\delta_{\mathbf{g}}(\mathbf{a}, \mathbf{x})_{ij} = \mathbf{g}(\mathbf{a}_{ij}^+, \mathbf{x}) - \mathbf{g}(\mathbf{a}_{ij}^-, \mathbf{x})$ is the change in network statistic that would occur if a_{ij} were changed from 0 to 1
 - ▶ where $\mathbf{a}_{ij}^+$ and $\mathbf{a}_{ij}^-$ represent the network $\mathbf{a}$ with dyad a_{ij} set to 1 or 0, respectively
- Then the equivalent ERGM specification is

$$\text{logit}\{\Pr_{\theta}(A_{ij} = 1 | \mathbf{A}_{ij}^C = \mathbf{a}_{ij}^C, \mathbf{X} = \mathbf{x})\} = \theta^k \delta_{\mathbf{g}}(\mathbf{a}, \mathbf{x})_{ij}$$

- ▶ where $\mathbf{A}_{ij}^C$ represents all dyads in $\mathbf{A}$ except A_{ij}
- **Interpretation of θ :** the change in conditional log-odds of the network associated with a one-unit increase in the corresponding component of $\mathbf{g}(\mathbf{a}, \mathbf{x})$ resulting from switching a particular dyad A_{ij} from 0 to 1 and leaving the rest of the network fixed at $\mathbf{A}_{ij}^C$

Dyadic Independence EGRMs

- **Dyadic independence term:** a component g of $\mathbf{g}$ in an ERGM for which the corresponding change statistic $\delta_g(\mathbf{a}, \mathbf{x})_{ij}$ can be calculated for any i, j without knowing $\mathbf{a}$
 - ▶ For example, if $g(\mathbf{a}, \mathbf{x}) = \sum_{i,j} a_{ij}(Z_i + Z_j)$ counts the number of edges touching treated nodes, then $\delta_g(\mathbf{a}, \mathbf{x})_{ij} = z_i + z_j$ doesn't depend on $\mathbf{a}$
- **Dyadic independence ERGM:** an ERGM with only dyadic independence terms
 - ▶ replace $\delta_g(\mathbf{a}, \mathbf{x})_{ij}$ with $\delta_g(\mathbf{x})_{ij}$ and write the model as

$$\text{logit}\{\Pr(A_{ij} = 1 | \mathbf{X} = \mathbf{x})\} = \theta \cdot \delta_g(\mathbf{x})_{ij}$$

- **Interpretation of θ :** the change in log-odds of the network associated with a one-unit increase in the corresponding component of $\mathbf{g}(\mathbf{a}, \mathbf{x})$ resulting from switching a particular dyad A_{ij} from 0 to 1

STERGM Model Formulation

- **Formation model** is an ERGM conditional on only adding edges:

$$\Pr_{\theta^+}(\mathbf{A}^{k+1} = \mathbf{a}^{k+1} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^+ \cdot \mathbf{g}^+(\mathbf{a}^{k+1}, \mathbf{x})\}}{\kappa\{\theta^+, \mathcal{A}^+(\mathbf{a}^k), \mathbf{x}\}}$$

► $\mathcal{A}^+(\mathbf{a})$: space of possible networks that can be formed by adding edges to $\mathbf{a}$

- **Persistence model** is an ERGM conditional on only removing edges:

$$\Pr_{\theta^-}(\mathbf{A}^{k-1} = \mathbf{a}^{k-1} | \mathbf{A}^k = \mathbf{a}^k, \mathbf{X} = \mathbf{x}) = \frac{\exp\{\theta^- \cdot \mathbf{g}^-(\mathbf{a}^{k-1}, \mathbf{x})\}}{\kappa\{\theta^-, \mathcal{A}^-(\mathbf{a}^k), \mathbf{x}\}}$$

► $\mathcal{A}^-(\mathbf{a})$: space of possible networks that can be formed by removing edges to $\mathbf{a}$

- A **STERGM** assumes the network at time $k + 1$ is then the result of applying the changes in $\mathbf{A}^{k+1}$ and $\mathbf{A}^{k-1}$ to $\mathbf{A}^k$:

$$\mathbf{A}^{k+1} = \underbrace{\mathbf{A}^k}_{\text{previous network}} \cup \underbrace{(\mathbf{A}^{k+1} - \mathbf{A}^k)}_{\text{new edges formed}} - \underbrace{(\mathbf{A}^k - \mathbf{A}^{k-1})}_{\text{old edges not persisting}}$$

Simulation Setup

- $n \in \{24, 48\}$ students were equally divided into two residence halls, each with six clusters of equal size
- Within each residence hall group, the clusters were randomized using a 50:50 allocation to either the intervention group or the control group
- Five pairs of students were randomly selected to be roommates, meaning they had contact with each other each week
- Baseline ($k = 1$) social contacts were simulated between each pair of students with probability 0.5
- Baseline infection statuses were simulated for each student with probability 0.5
- Social networks were simulated over the remaining $\tau \in \{5, 10\}$ weeks according to a STERGM with both formation and persistence models including edge count, intervention assignment Z , infection status Y , $Z \times Y$ interaction, and an offset to force constant edges between roommates
- Infection statuses were simulated for each student i at each week $k \in \{2, \dots, \tau\}$ with probability $\Pr(Y_i^k = 1 | \mathbf{A}^{k-1}, \mathbf{Y}^{k-1}) = g(\sum_{j=1}^n A_{ij}^{k-1} Y_j^{k-1})$, where $\sum_{j=1}^n A_{ij}^{k-1} Y_j^{k-1}$ is the number of infected contacts at the previous week, and $g : [0, \infty) \rightarrow [0, 1]$ is a non-decreasing function

Simulation Setup

Three scenarios:

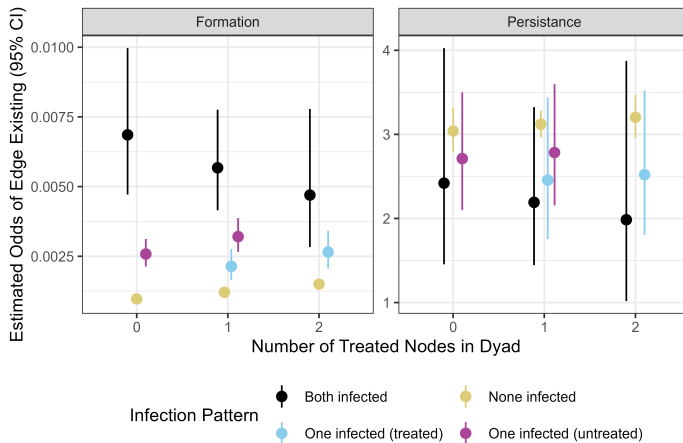
Null Hypothesis	Formation Model Parameters θ^+	Persistence Model Parameters θ^-	Infection Probability Function g
$H_0^\sharp$	$(-0.5, 0, 0, 0)$	$(-0.5, 0, 0, 0)$	$g(s) = 0.5$
$\overline{H}_0^A \cap H_0^Y$	$(-0.2, 0, 0, -1)$	$(-0.2, 0, 0, -1)$	$g(s) = 0.5$
$\overline{H}_0^A \cap \overline{H}_0^Y$	$(-0.2, 0, 0, -1)$	$(-0.2, 0, 0, -1)$	$g(s)$ increasing in s

Table: Data generating process for the simulation study. The null hypothesis refers to the null hypothesis that is true under the data generating process, formation model parameters are $\theta^+ = (\theta_{\text{edges}}^+, \theta_Z^+, \theta_Y^+, \theta_{ZY}^+)$, persistence model parameters are $\theta^- = (\theta_{\text{edges}}^-, \theta_Z^-, \theta_Y^-, \theta_{ZY}^-)$, and the infection probability function g provides a unit's probability of being infected at week k given their number of infected neighbors at week $k - 1$.

Three p-values:

- $\rho_B^\sharp$: testing the sharp null
- ρ_B^Y : testing H_0^Y using known q
- $\hat{\rho}_B^Y$: testing H_0^Y using estimated q

eX-FLU: STERGM Results



Type I Error Control

- **Proposition 2.1:** Let T_N be a test statistic with CDF F_N under H_0 .
 - ① Under H_0 , $F_N(T_N)$ stochastically dominates a $\text{Uniform}(0, 1)$ distribution for any N .
 - ② If the test statistic T_N has a continuous limiting distribution, then $F_N(T_N) \rightarrow^d \text{Uniform}(0, 1)$.
- **Corollary 2.2:**
 - ① Under $H_0^\sharp$, the sharp null p-value $\rho_N^\sharp$ stochastically dominates a $\text{Uniform}(0, 1)$ distribution for any N .
 - ② Under H_0^Y , the oracle p-value ρ_N^Y stochastically dominates a $\text{Uniform}(0, 1)$ distribution for any N .
 - ③ If the test statistic T_N has a continuous limiting distribution, then $\rho_N^\sharp \rightarrow^d \text{Uniform}(0, 1)$ under $H_0^\sharp$ and $\rho_N^Y \rightarrow^d \text{Uniform}(0, 1)$ under H_0^Y .

Type I Error Control

- **Proposition 2.3:** Let $q(\mathbf{a}, \mathbf{z}, \theta) \equiv \Pr\{\mathbf{A}(\mathbf{z}) = \mathbf{a}; \theta\}$ denote the PMF of the distribution of stochastic potential networks $\mathbf{A}(\mathbf{z})$ at parameter value θ , let $\hat{\theta}_N$ denote the estimator of θ , and let $F_N(\cdot; \theta)$ denote the CDF of the test statistic T_N at θ , with limiting CDF $F(\cdot; \theta)$. Let θ_0 denote the true value of θ . Assume the following:

(A1) $\hat{\theta}_N \rightarrow^P \theta_0$

(A2) $F(t; \theta_0)$ is continuous in t on $\mathbb{R}$

(A3) there exists a $\delta_0 > 0$ such that

$$\sup_{\theta \in B_{\delta_0}(\theta_0)} \sup_{t \in \mathbb{R}} |F_N(t; \theta) - F(t; \theta)| \rightarrow 0$$

(A4) $F(t; \theta)$ is continuous in θ at θ_0 uniformly in t , i.e.,

$$\lim_{\theta \rightarrow \theta_0} \sup_{t \in \mathbb{R}} |F(t; \theta) - F(t; \theta_0)| = 0$$

Then the plug-in p-value $\hat{\rho}_N^Y$ converges in distribution to $\text{Uniform}(0, 1)$.

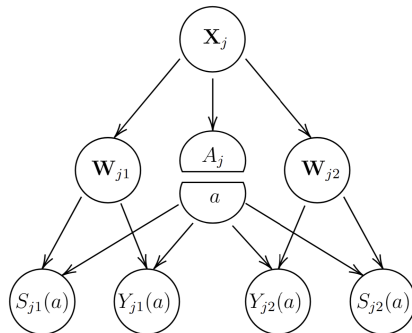
Type I Error Control

- **Proposition 2.4:** Let $\rho_N = F_N(T_N)$ for test statistic T_N and CDF F_N (not necessarily the true CDF of T_N). Let $T_N^* = h_N(T_N)$ for a sequence of deterministic, strictly increasing functions h_N . Define $F_N^*(t) = F_N\{h_N^{-1}(t)\}$, the (not necessarily true) CDF of the transformed test statistic, and let $\rho_N^* = F_N^*(T_N^*)$. Then $\rho_N^* = \rho_N$.
- **Corollary 2.5:** Let h_N be a sequence of deterministic, strictly increasing functions.
 - 1 If the hypotheses of Proposition 2.1 are met for a test statistic T_N , then the results also hold for $T_N^* = h_N(T_N)$.
 - 2 If the hypotheses of Proposition 2.3 are met for $T_N, F_N(\cdot; \theta)$, then the results also hold for $T_N^* = h_N(T_N), F_N^*(\cdot; \theta) = F_N\{h_N^{-1}(\cdot); \theta\}$.

Working Assumptions for Identifiability

For all clusters j, j' , individuals i and treatment a :

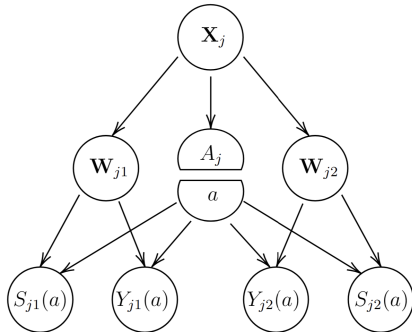
- causal consistency: $Y_{ji} = Y_{ji}(a)$ and $S_{ji} = S_{ji}(a)$ when $A_j = a$
- positivity of treatment assignment:
 $\Pr(A_j = a | \mathbf{X}_j = \mathbf{x}) > 0$ for all $\mathbf{x}$ with $\Pr(\mathbf{X}_j = \mathbf{x}) > 0$
- positivity of response: $\Pr(S_{ji} = 1 | A_j = a, \mathbf{X}_j = \mathbf{x}, \mathbf{W}_{ji} = \mathbf{w}) > 0$
for all $\mathbf{x}, \mathbf{w}$ with
 $\Pr(\mathbf{X}_j = \mathbf{x}, \mathbf{W}_{ji} = \mathbf{w}) > 0$
- independent clusters:
$$\left[A_j, \mathbf{X}_j, \{R_{ji}, S_{ji}(a)\}_{i=1}^{N_j}, \mathbf{Y}_j(a), Q_{ji} \right] \perp\!\!\!\perp \left[A_{j'}, \mathbf{X}_{j'}, \{R_{j'i}, S_{j'i}(a)\}_{i=1}^{N_{j'}}, \mathbf{Y}_{j'}(a), Q_{j'i} \right]$$



Working Assumptions for Identifiability

For all clusters j, j' , individuals i and treatment a :

- conditional exchangeability: $Y_{ji}(a) \perp\!\!\!\perp A_j | \mathbf{X}_j, \mathbf{W}_{j,i}$
- conditional independence of potential outcome and responses:
 $Y_{ji}(a) \perp\!\!\!\perp S_{j,i}(a) | A_j, \mathbf{X}_j, \mathbf{W}_{j,i}$
- cluster-level randomization: $A_j \perp\!\!\!\perp (\mathbf{W}_{j1}, \dots, \mathbf{W}_{jN_j}) | \mathbf{X}_j$
- independent random sampling:
 $R_{ji} \perp\!\!\!\perp [A_j, \mathbf{X}_j, \{S_{ji}(a)\}_{i=1}^{N_j}, \mathbf{Y}_j(a), Q_{ji}]$
- independent random sampling of auxiliary data:
 $Q_{ji} \perp\!\!\!\perp [A_j, \mathbf{X}_j, \{R_{ji}, S_{ji}(a)\}_{i=1}^{N_j}, \mathbf{Y}_j(a)]$



Estimators: G-Formula

Define the conditional mean outcome among trial responders with cluster-level covariates $\mathbf{x}$ and individual-level covariates $\mathbf{w}$, and under treatment assignment a by

$$\mu(\mathbf{x}, \mathbf{w}) = \text{E} (Y_{ji} | A_j = a, R_{ji} = 1, S_{ji} = 1, \mathbf{X}_j = \mathbf{x}, \mathbf{W}_{ji} = \mathbf{w}), \quad (1)$$

The g-formula estimator averages $\mu(\mathbf{x}, \mathbf{w})$ over the empirical distribution of $(\mathbf{X}_j, \mathbf{W}_{ji})$ in the auxiliary data:

$$\hat{\eta}^{GF}(a) = \frac{1}{m} \sum_{j=1}^m \frac{1}{\tilde{n}_j} \sum_{i=1}^{N_j} Q_{ji} \mu(\mathbf{X}_j, \mathbf{W}_{ji}). \quad (2)$$